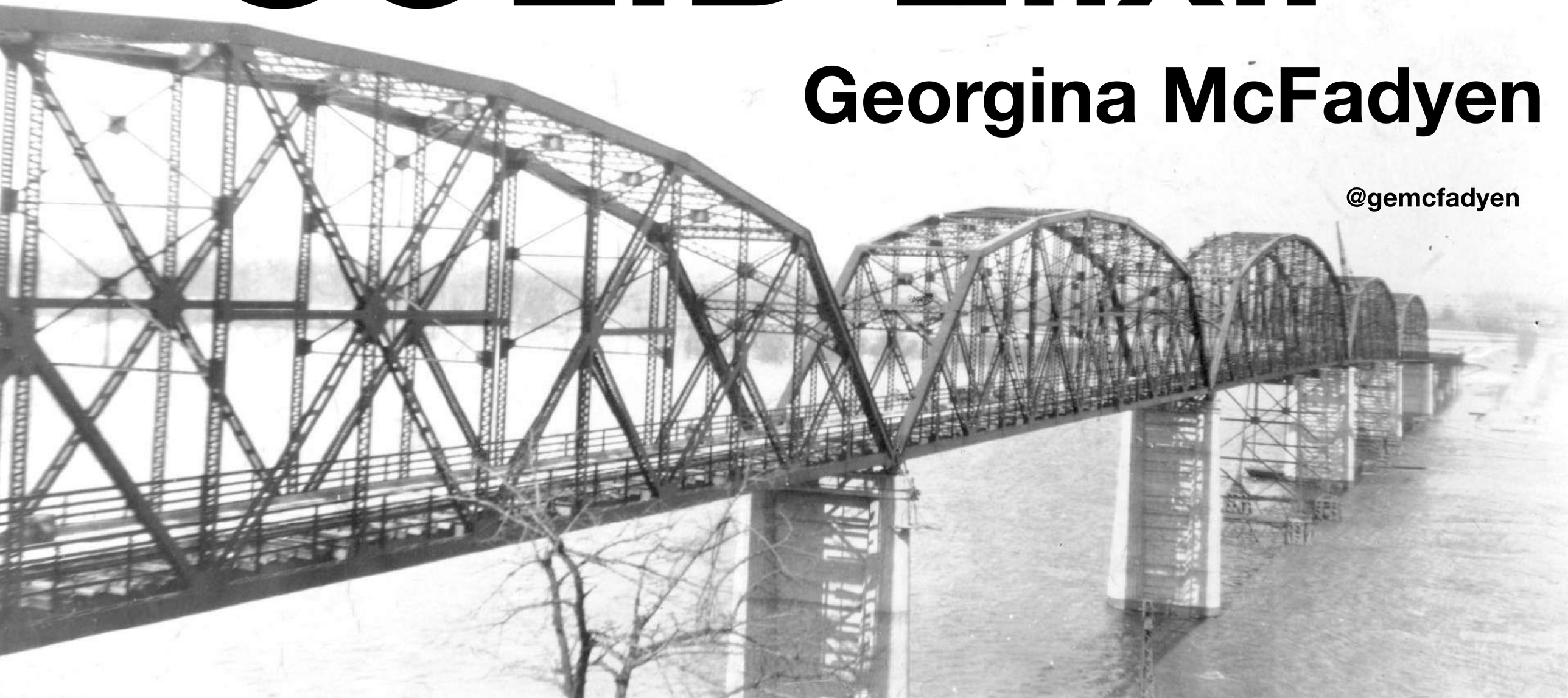
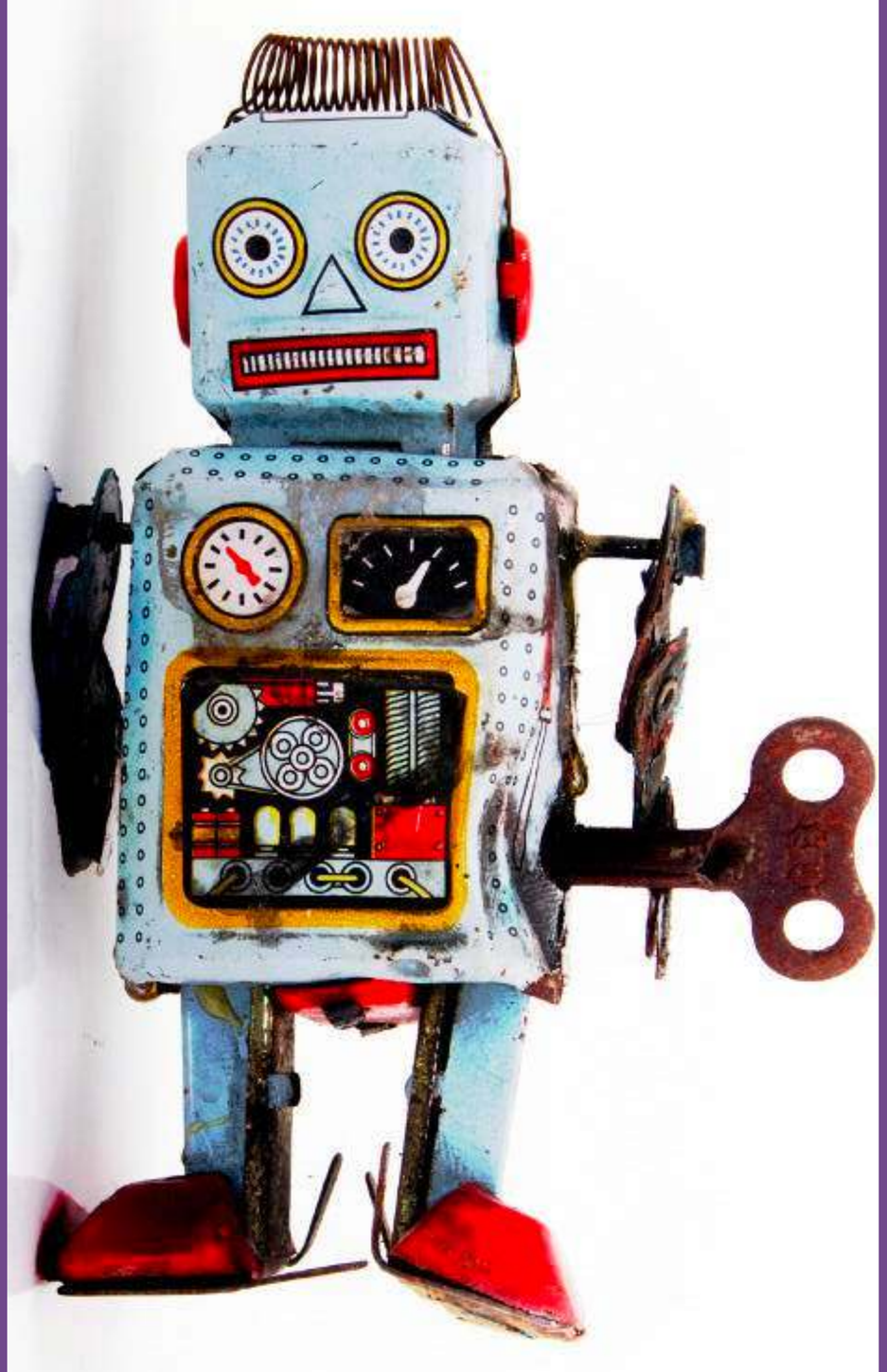
SOLID Elixir

Georgina McFadyen

@gemcfadyen



Hi, I’m Georgina



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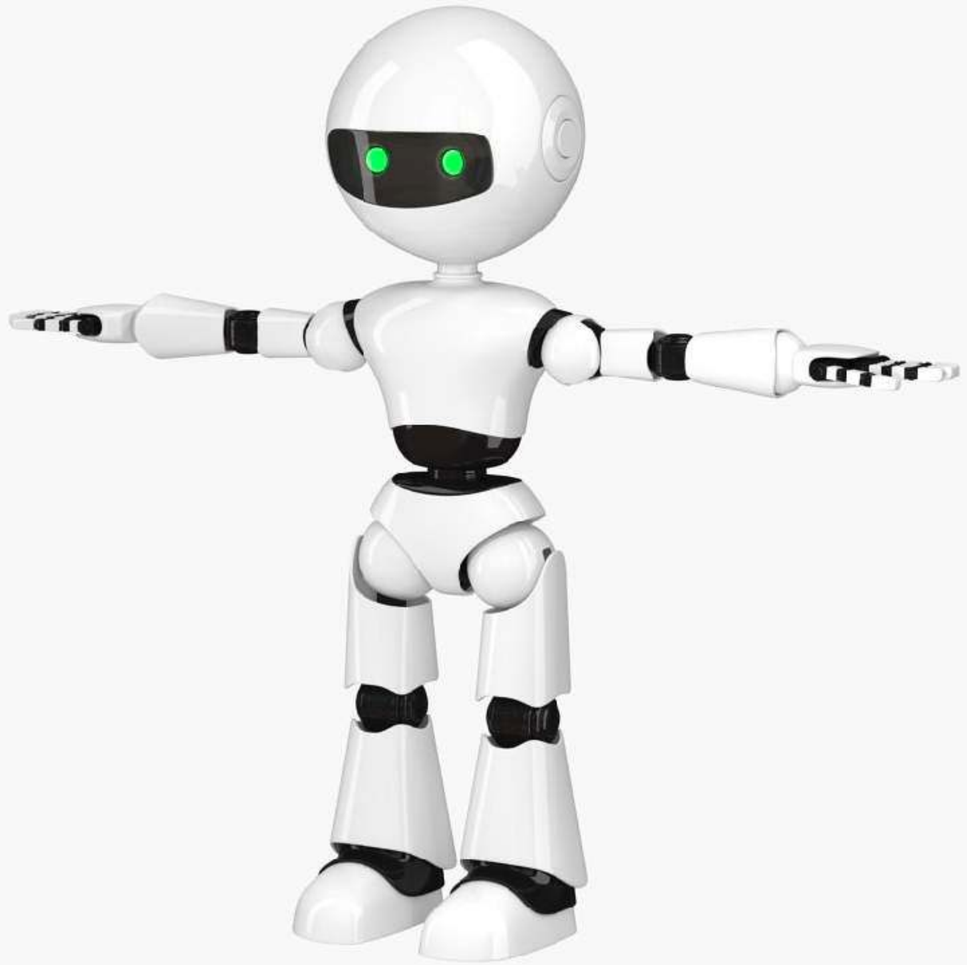

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What are the SOLID principles?

“In Object-Oriented programming, the term SOLID is a mnemonic acronym for five design principle intended to make software design more understandable flexible and maintainable .”

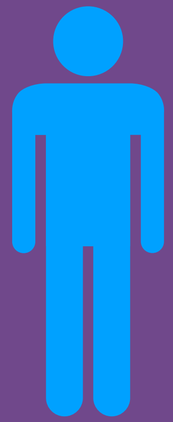
– Wikipedia
https://en.wikipedia.org/wiki/SOLID_(object-oriented_design)

Holiday Booking Domain

https://github.com/gemcfadyen/solid_elixir


Select your accommodation preferences
Hotel
Apartment
Guest House
Bedrooms
1
2
3
4
Catering
Self-Catering
All Inclusive
Parking
None
On Street
Secure Garage
Search

GET users preferences



Login

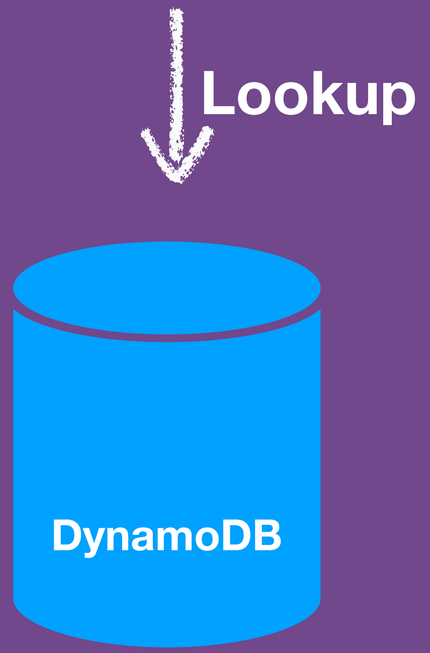
GET users preferences

Login

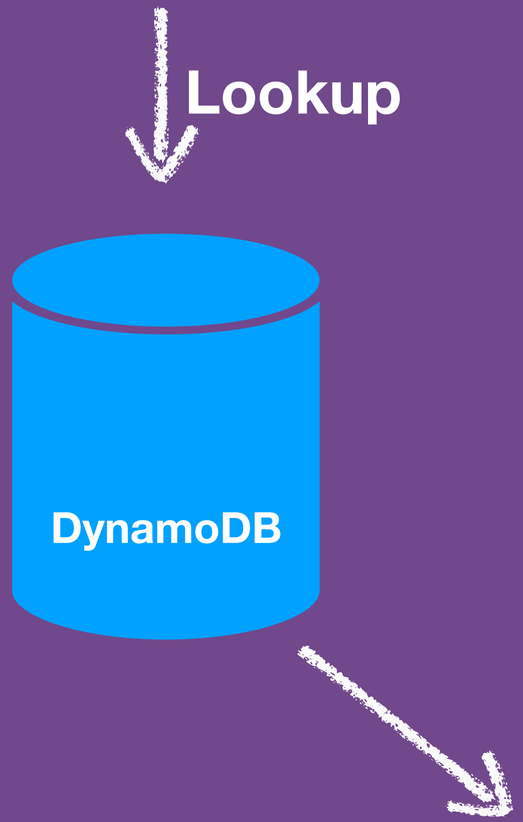
id of customer

GET /customers/{id}

GET /customers/{id}



GET /customers/{id}




```
%{"id" => "uuid-1",
    "lastModified" => 1519928745635,
    "preferences" =>
        %{"accommodation" =>
            %{"apartment" => %{
                "catering" => "self_catering",
                "bedrooms" => 2,
                 "parking" => "secure"
                },
                "hotel" => %{
                    "catering" => "self_catering",
                    "bedrooms" => 1,
                    "parking" => "secure"
                 }
            }
    }
}
```

GET /customers/{id}

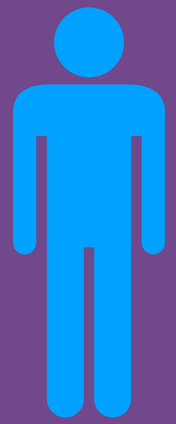
```
%{"id" => "uuid-1",
    "lastModified" => 1519928745635,
    "preferences" =>
        %{"accommodation" =>
            %{"apartment" => %{
                "catering" => "self_catering",
                "bedrooms" => 2,
                 "parking" => "secure"
                },
                "hotel" => %{
                   "catering" => "self_catering",
                   "bedrooms" => 1,
                   "parking" => "secure"
                }
        }
    }
}
```

Internal Struct Representation

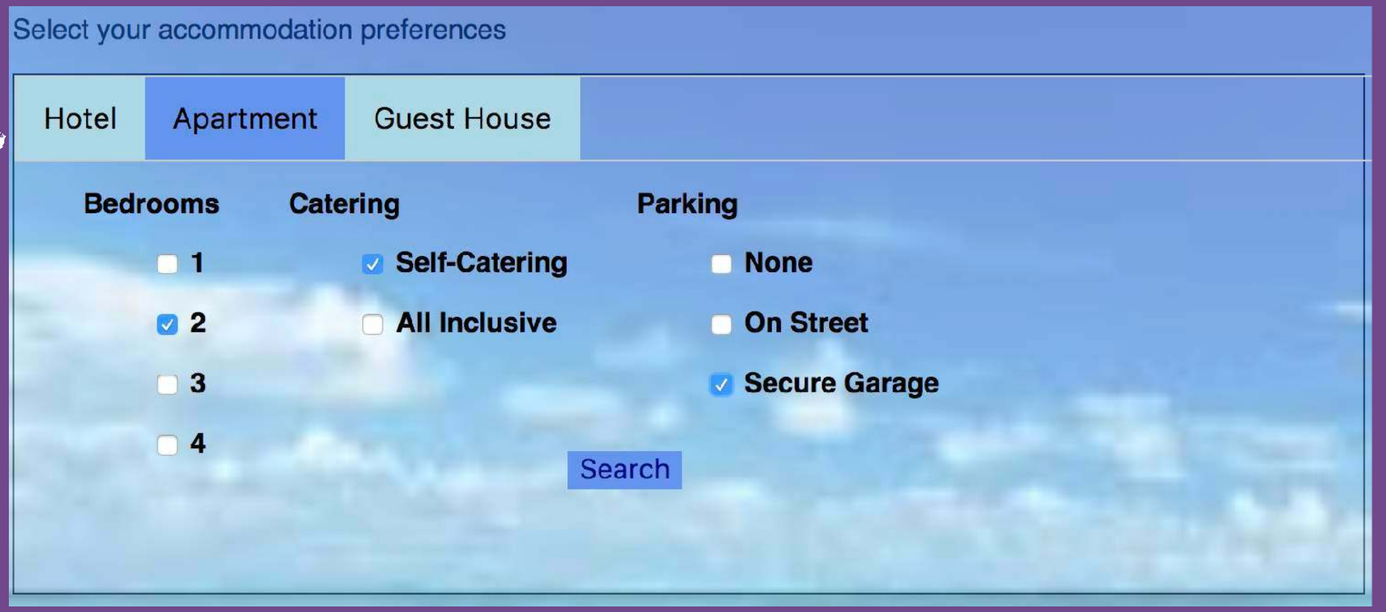
```
{
  "id":"uuid-1",
  "lastModified": "2018-02-21T12:05:15Z",
  "preferences": {
    "accommodation": {
        "apartment": {
                "catering": "self_catering",
                "bedrooms": 2,
                 "parking": "secure"
                },
                "hotel": {
                   "catering": "self_catering",
                   "bedrooms": 1,
                   "parking": "secure"
                }
        }
    }
}
```

JSON Representation

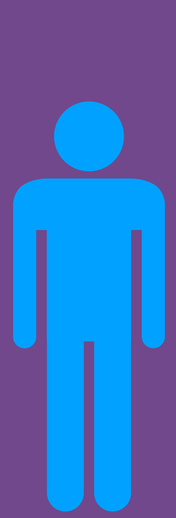
POST users preferences



Login

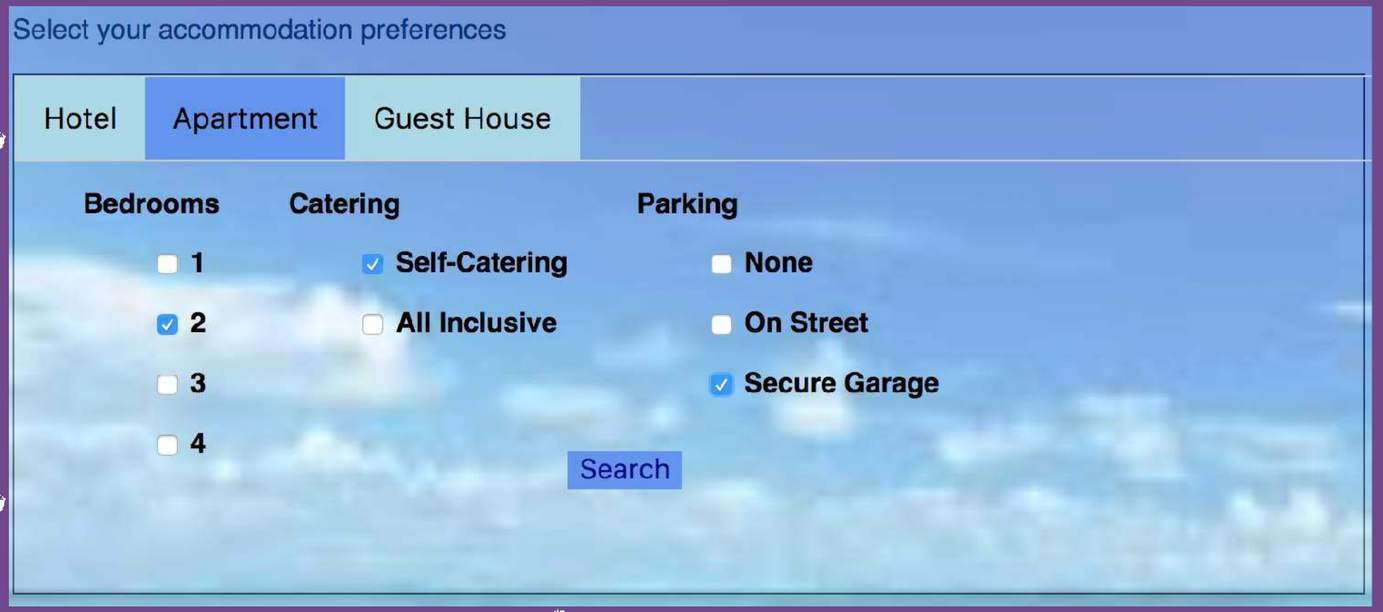


POST /customers



Login

Search



POST /customers

POST /customers

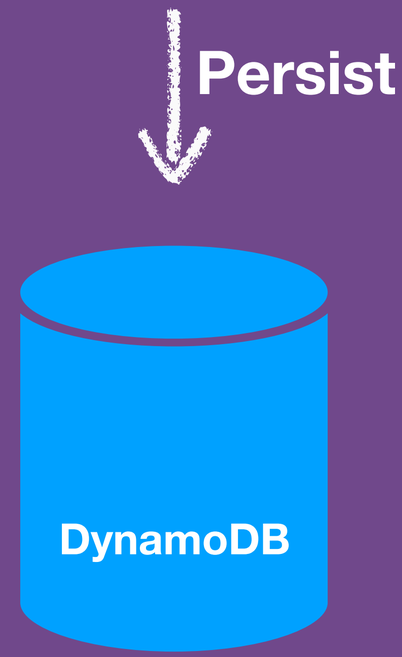
```
{
  "id":"uuid-1",
  "lastModified": "2018-02-21T12:05:15Z",
  "preferences": {
    "accommodation": {
        "apartment": {
              "catering": "self_catering",
              "bedrooms": 2,
               "parking": "secure"
               },
              "hotel": {
                 "catering": "self_catering",
                 "bedrooms": 1,
                 "parking": "secure"
              }
        }
    }
}
```

JSON Representation

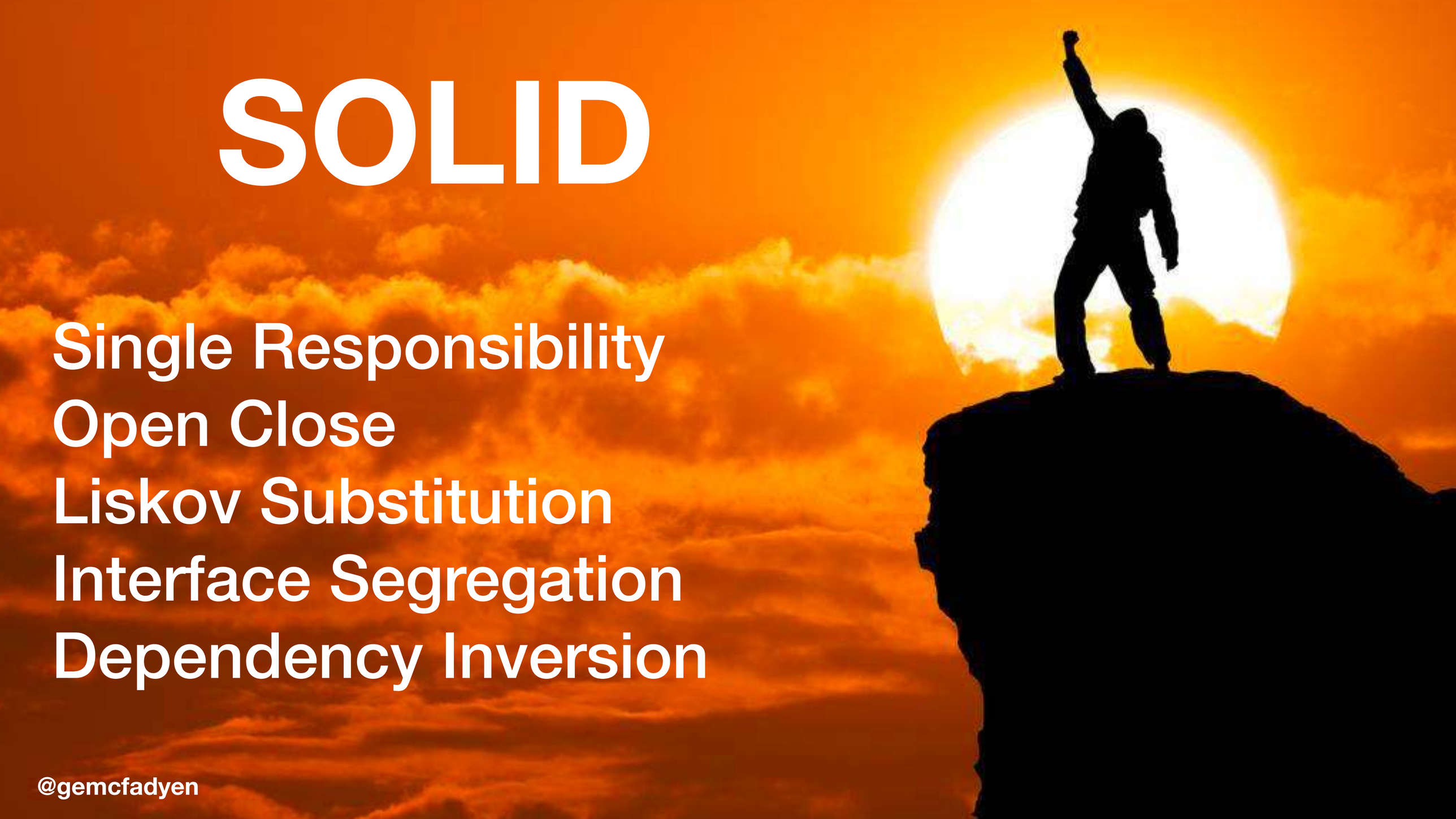
```
%{"id" => "uuid-1",
    "lastModified" => 1519928745635,
    "preferences" =>
       %{"accommodation" =>
           %{"apartment" => %{
              "catering" => "self_catering",
              "bedrooms" => 2,
               "parking" => "secure"
               },
              "hotel" => %{
                 "catering" => "self_catering",
                 "bedrooms" => 1,
                 "parking" => "secure"
               }
        }
    }
}
```

Internal Struct Representation

POST /customers



SOLID



Single Responsibility
Open Close
Liskov Substitution
Interface Segregation
Dependency Inversion

@gemcfadyen


Single Responsibility
SOLID
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“The single responsibility principle states that every module or class should have responsibility over a single part of the functionality provided by the software...”

– Wikipedia
https://en.wikipedia.org/wiki/Single_responsibility_principle

```elixir
defmodule Controllers.PreferenceController do

  def get_preferences(customer_id) do
    ExAws.Dynamo.get_item("customer-preferences", %{id: customer_id})
    |> ExAws.request
    |> decode
  end
  defp decode({:ok, response}) when map_size(re
   {:error, :not_found}
  end
  defp decode({:ok, response}) do
   valid_consent_user = ExAws.Dynamo.decode_item(response,
                                as: Schema.Database.CustomerPreferencesRow)
   |> format_response()

   {:ok, valid_consent_user}
  end
  defp decode({:error, response}) do
   {:error, response}
  end

  def format_response(raw_response) do
   pretty_response = raw_response
   # Data formatting to create a nice response for calling system
   pretty_response
  end
end
```

Fetch User Preferences

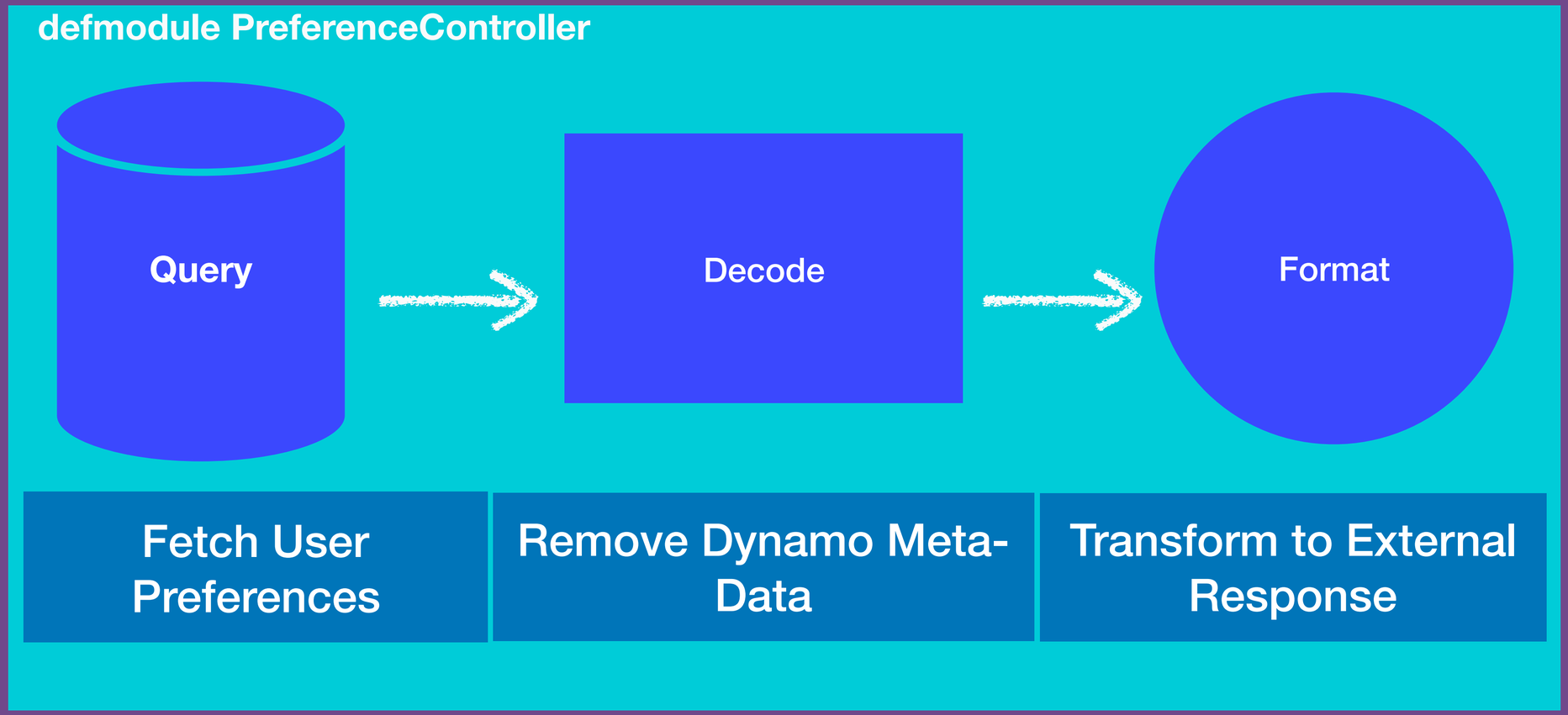
```elixir
defmodule Controllers.PreferenceController do

  def get_preferences(customer_id) do
    ExAws.Dynamo.get_item("customer-preferences", %{id: customer_id})
    |> ExAws.request
    |> decode
  end
  defp decode({:ok, response}) when map_size(response) == 0 do
    {:error, :not_found}
  end
  defp decode({:ok, response}) do
    valid_consent_user = ExAws.Dynamo.decode_item(response,
                                                  as: Schema.Database.CustomerPreferencesRow)
    |> format_response()

    {:ok, valid_consent_user}
  end
  defp decode({:error, response}) do
    {:error, response}
  end

  def format_response(raw_response) do
    pretty_response = raw_response
    # Data formatting to create a nice response for calling system
    pretty_response
  end
end
```

Remove Dynamo Meta-Data

```elixir
defmodule Controllers.PreferenceController do

  def get_preferences(customer_id) do
    ExAws.Dynamo.get_item("customer-preferences", %{id: customer_id})
    |> ExAws.request
    |> decode
  end
  defp decode({:ok, response}) when map_size(response) == 0 do
    {:error, :not_found}
  end
  defp decode({:ok, response}) do
    valid_consent_user = ExAws.Dynamo.decode_item(response,
                                  as: Schema.Database.CustomerPreferencesRow)
    |> format_response()

    {:ok, valid_consent_user}
  end
  defp decode({:error, response}) do
    {:error, response}
  end

  def format_response(raw_response) do
    pretty_response = raw_response
    # Data formatting to create a nice response for calling system
    pretty_response
  end
end
```

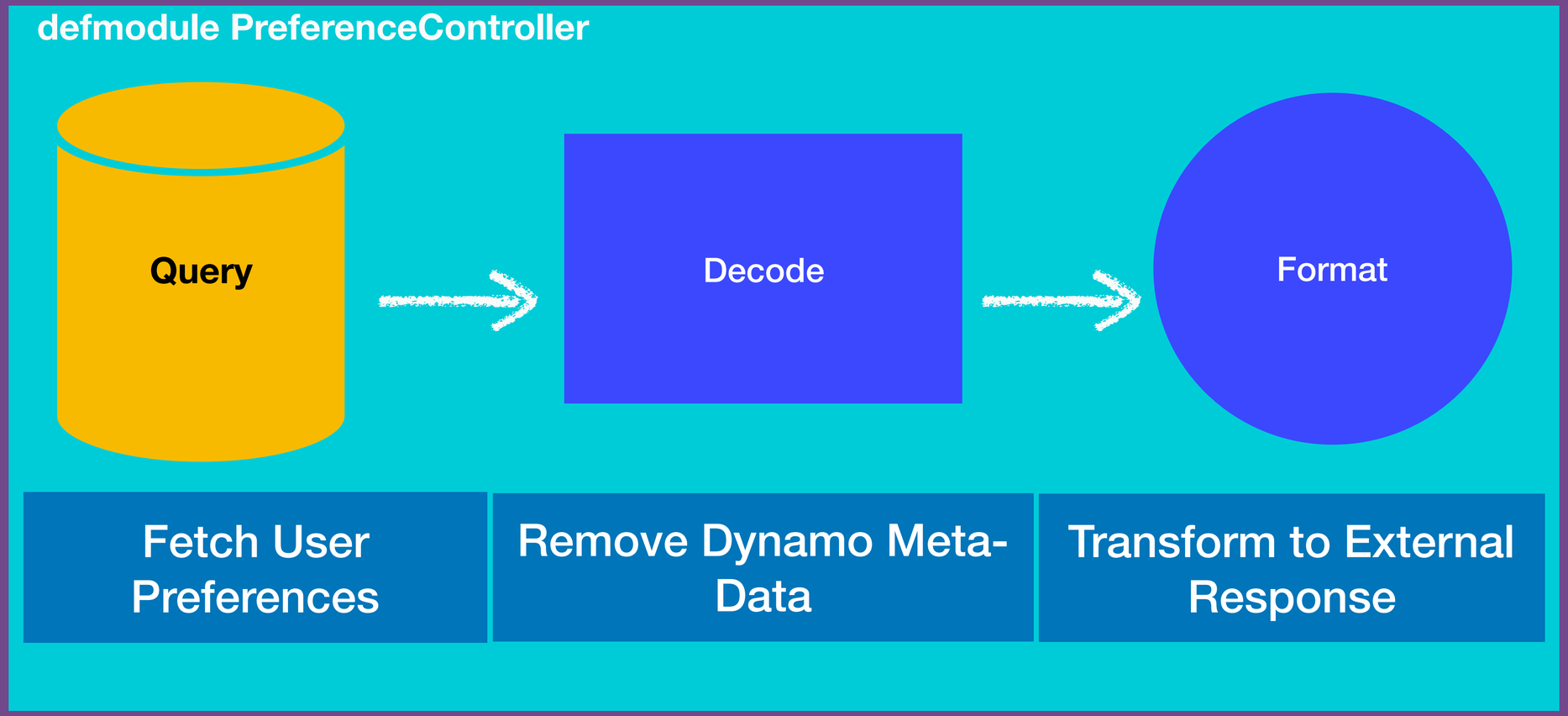
Transform to External Response

GET
Controller Responsibilities



Can any of these steps change independently?

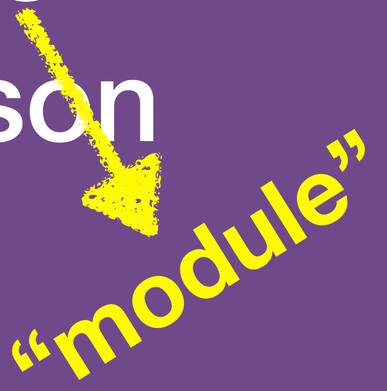
What if we changed the conditions on which we query?



“Robert C. Martin expresses the principle as, "A class should have only one reason to change.”

– Wikipedia
https://en.wikipedia.org/wiki/Single_responsibility_principle

“Robert C. Martin expresses the principle as, "A class should have only one reason to change.”



– Wikipedia
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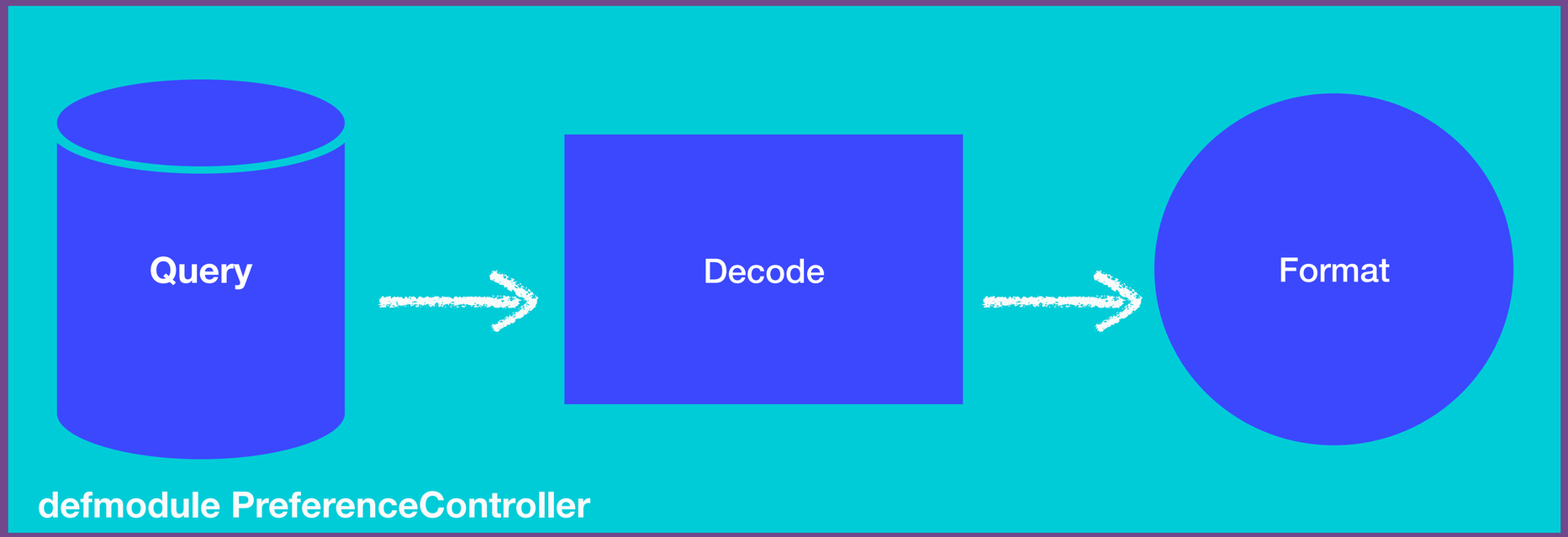
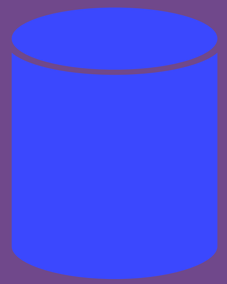
Controller Responsibilities



Query Database

```
defmodule Core.Preferences.GetCustomerPreferences do

  def for(customer_id) do
   ExAws.Dynamo.get_item("customer-preferences",
                         %{id: customer_id})
   |> ExAws.request
  end
end
```



Decode Database Response

```elixir
defmodule Core.Preferences.Decode do

  def decode({:ok, response}) when map_size(response) == 0 do
   {:error, :not_found}
  end
  defp decode({:ok, response}) do
   valid_preferences = ExAws.Dynamo.decode_item(response,
                       as: Schema.Database.CustomerPreferencesRow)

   {:ok, valid_prefereneces}
  end
  defp decode({:error, response}) do
   {:error, response}
  end
end
```

Decode Database Response

```
valid_preferences = ExAws.Dynamo.decode_item(response,
                    as: Schema.Database.CustomerPreferencesRow)
```

Format to Json Response

```elixir
defmodule Core.Schema.External.Response do
  def format({:ok, database_response}) do
    database_response
    |> format_dates()
    |> group_accomodation_preferences()
  end

  defp format_dates(raw_data) do
    ..
  end

  defp group_accomodation_preferences(raw_data) do
    ...
  end
end
```

Controller

```elixir
defmodule Controllers.PreferenceController do

  def get_preferences(customer_id) do
   Core.Preferences.GetCustomerPreferences.for(customer_id)
   |> Core.Schema.External.Response.format()
  end
end
```

Benefits of Single Responsibility

SOLID

- ✅ Focused unit tests
- ✅ Fewer higher level tests
- ✅ Less dependencies per test
- ✅ Similar functionality grouped

Trade Off

❌ Ensure each module justifies it’s existence


Open Close
SOLID
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“Software entities (classes, modules, functions, etc.) should be open for extension, but closed for modification…An entity can allow its behaviour to be extended without modifying its source code”

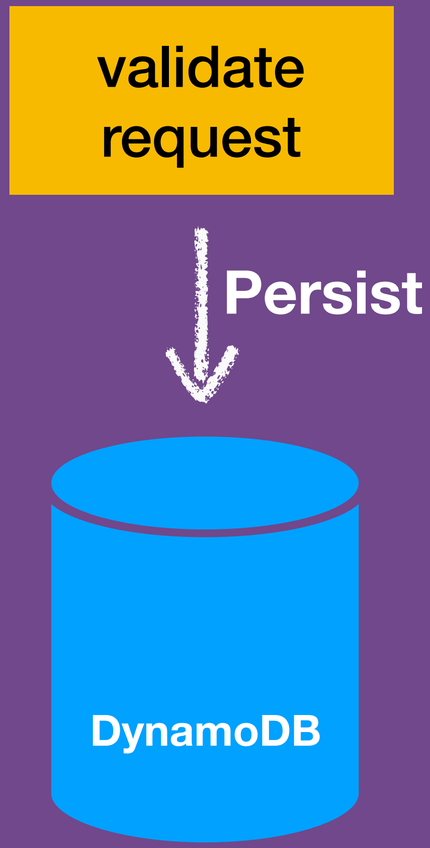
– Wikipedia
https://en.wikipedia.org/wiki/Open/closed_principle

Open Closed

Ability to add new code without modifying any existing code.

POST /customers

POST /customers



Request Validation

? Is query parameter a valid uuid?

? Does body contain preferences section?

? Does body contain accommodation section?

```elixir
def call(conn, _opts) do
 ...
 with {:ok, customer_id} <-
      Validation.RequestParameterValidation
                .has_valid_parameters(conn.path_params),
           :ok <-
             Validation.PreferencesValidation
                           .has_preferences(body),
           :ok <-
            Validation.AccommodationValidation
                      .has_accommodation_preferences(body) do

          ...
```

validate different parts of request

Save Preferences

```
18 valid_request = Schema.Http.Request.new!(body,
19                                          customer_id,
20                                          body["version"])
21
   response =
     Controllers.SavePreferenceController.save_preferences(customer_id,
                                                            valid_request)
   send_resp(conn, 201, Poison.encode!(response))
```

Handle Errors

```
else
  {:error, :invalid_url_params} ->
     send_resp(conn, 400, Poison.encode!(%{error:
                  "URL Parameters need to be alpha numeric"}))
  {:error, :no_preferences} ->
     send_resp(conn, 400, Poison.encode!(%{error:
                   "Preferences section missing"}))
  {:error, :no_accommodation}  ->
     send_resp(conn, 400, Poison.encode!(%{error:
                    "Accommodation section missing"}))
end
```

Add new modules which checks the new rule

Add new modules which checks the new rule

Handle error case

- Don’t want to open up the Plug module each time a new validation rule is added
- Don’t want to have to test all the rule permutations at this high level

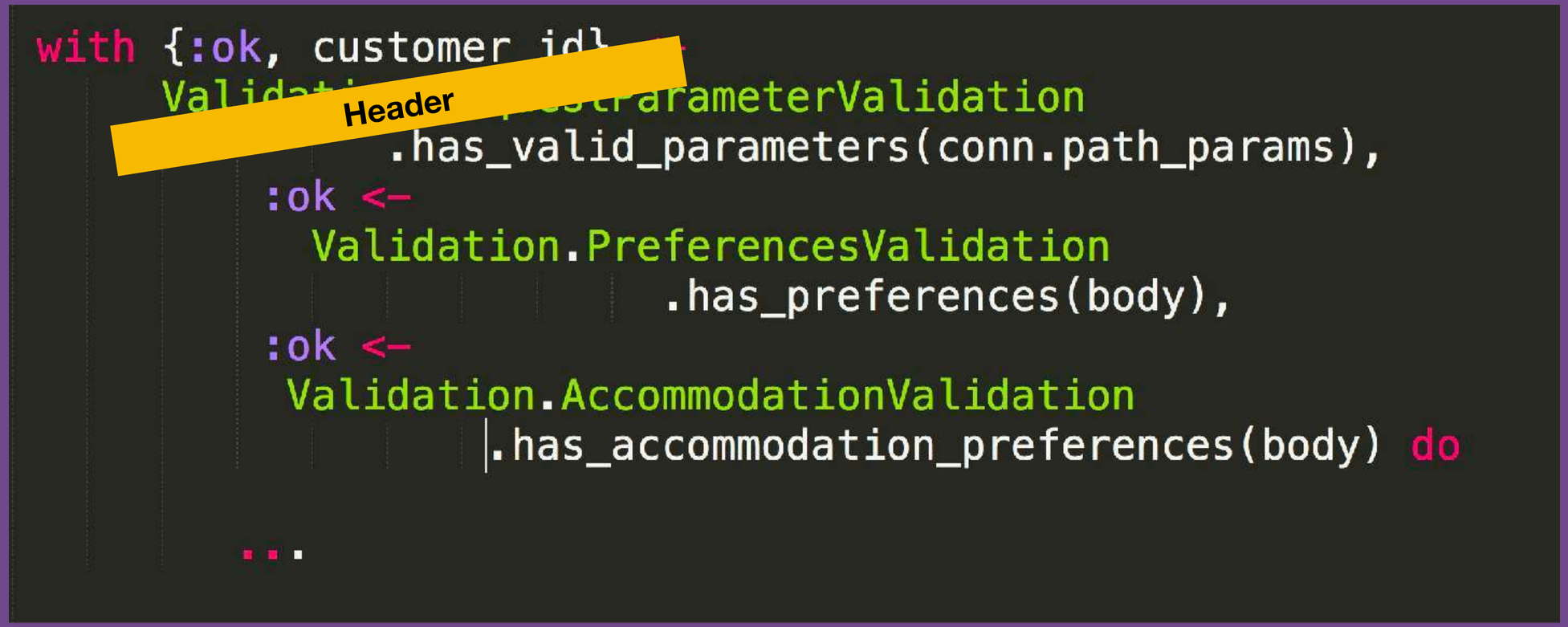
- Do want the ability to plug in new validation rules
- Do want to configure one rule for test env
- Do want to configure all rules for dev/prod env

Categorise Validations

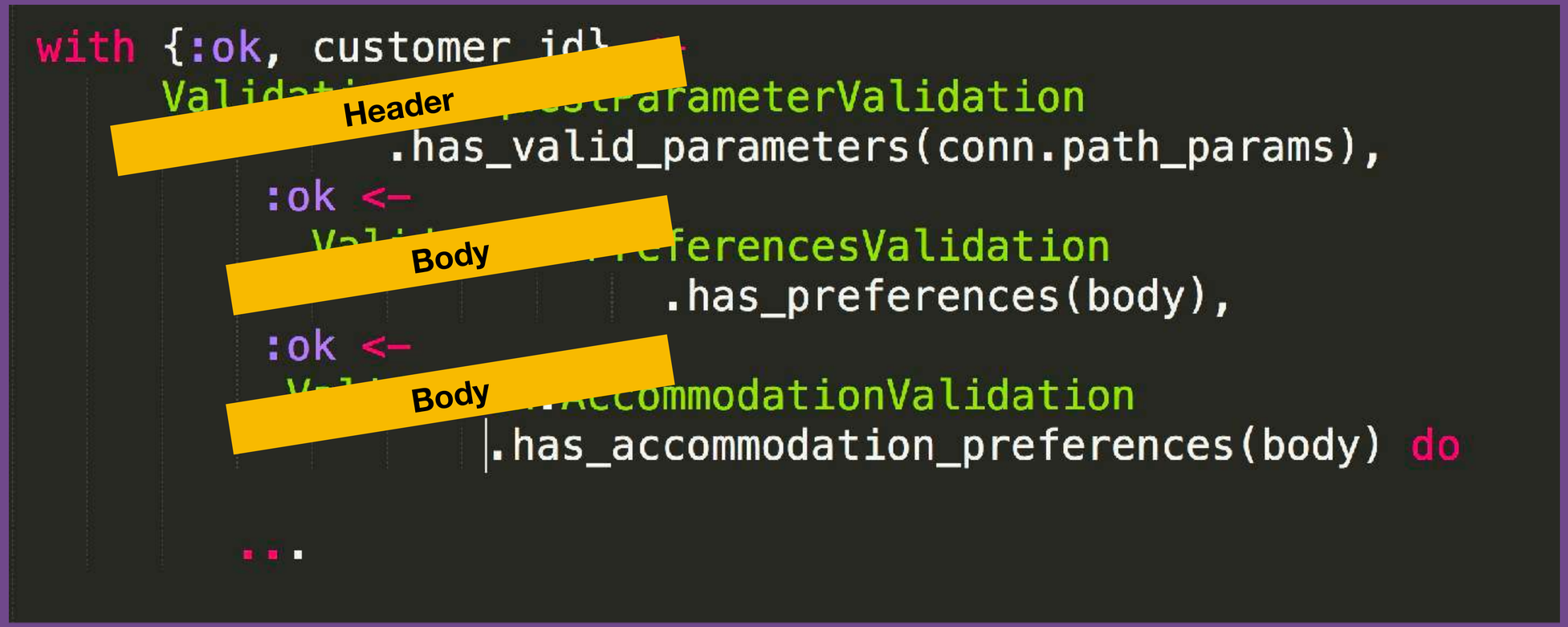
```
with {:ok, customer_id} <-
       Validation.RequestParameterValidation
                 .has_valid_parameters(conn.path_params),
           :ok <-
             Validation.PreferencesValidation
                           .has_preferences(body),
           :ok <-
            Validation.AccommodationValidation
                     .has_accommodation_preferences(body) do

          ...
```

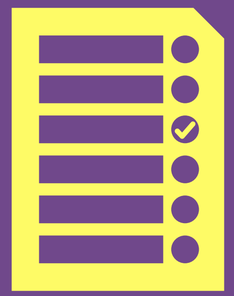
Categorise Validations



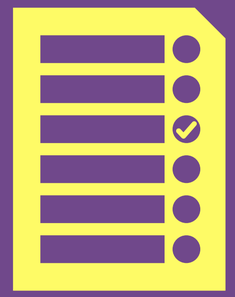
Categorise Validations



Define Validation Rules



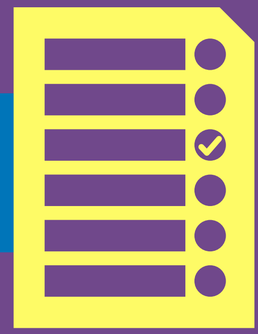
[HeaderRule1, HeaderRule2, HeaderRule3…]



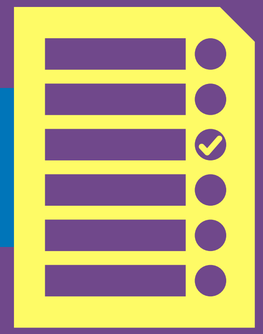
[BodyRule1, BodyRule2, BodyRule3…]

Categorise Validations

def module Validation.HeaderValidator



def module Validation.BodyValidator



```
3
4       parameter_validators =
            [&Validation.RequestParameterValidator.
                                    has_valid_parameters/1,
             &Validation.CustomerRestriction.unrestricted/1]
```

Configure Rules

```elixir
defp validate_all_rules([rule|tail], params) do
  {status, data} = rule.(params)

  if status == :error do
    {:error, data}
  else
    validate_all_rules(tail, params)
  end
end
```

Traverse and Execute Rules

Save Customer

Validation Per Category

```
with {:ok, customer_id} <-
               Validation.HeaderValidator.validate(path_params),
     {:ok, valid_body} <-
               Validation.BodyValidator.validate(body) do
```

Save Customer

```
with {:ok, customer_id} <-
               Validation.HeaderValidator.validate(path_params),
     {:ok, valid_body} <-
               Validation.BodyValidator.validate(body) do
```

```
parameter_validators =
    [&Validation.RequestParameterValidator.
                                has_valid_parameters/1,
     &Validation.CustomerRestriction.unrestricted/1]
```

Behaviours

Behaviours

“Behaviours provide a way to:

- define a set of functions that have to be implemented by a module
- ensure that a module implements all of the functions in that set.”

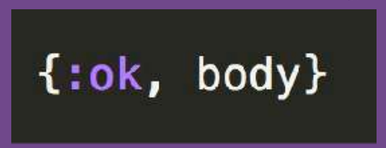
https://elixir-lang.org/getting-started/typespecs-and-behaviours.html#behaviours

Validation Rule Behaviour

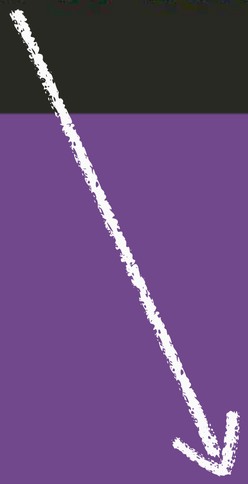
```
defmodule Validation do
  @callback is_valid(map)  ::   {:ok, term}
                              | {:error, {atom, reason :: String.t}}
end
```

Validation Rule Behaviour

```
defmodule Validation do
  @callback is_valid(map)   ::    {:ok, term}
                                | {:error, {atom, reason :: String.t}}
end
```



```
{:ok, body}
```



```
{:error, {:no_accommodation, "Accommodation preferences missing"}}
```

Implement the Behaviour

Implement Behaviour

```elixir
defmodule Validation.Body.ValidationRule do
  @behaviour Validation

  def is_valid(request_body) do
    ...
  end
end
```

Implement Behaviour

```elixir
defmodule Validation.Body.ValidationRule do
  @behaviour Validation

  def is_valid(request_body) do
    ...
  end
end
```

```
parameter_validators =
    [&Validation.RequestParameterValidator.
                                has_valid_parameters/1,
     &Validation.CustomerRestriction.unrestricted/1]
```

```
parameter_validators =
  [Validation.RequestParameterValidator,
   Validation.CustomerRestriction]
```

Modules implement Behaviour

```
{status, data} = rule.(params)
```

```
{status, data} = rule.is_valid(params)
```


Execute Contract

Configure the Rules

```
config :customer_preference_api,
  header_validators: [Validation.RequestParameterValidator,
                      Validation.CustomerRestriction
  ],
  body_validators: [Validation.Body.PreferencesValidation,
                    Validation.Body.AccommodationValidation]

```

Configure the Rules

```
7 header_validators =
8   Application.get_env(:customer_preference_api, :header_validators)
  body_validators =
    Application.get_env(:customer_preference_api, :body_validators)
```

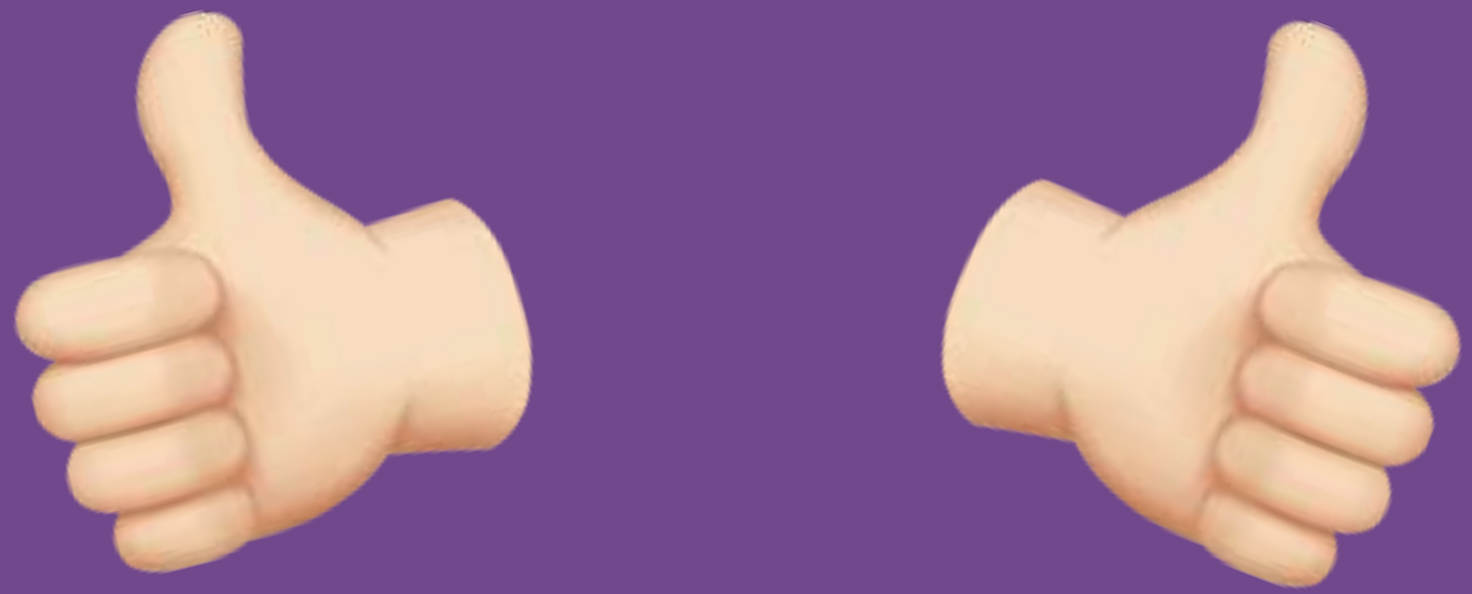
To Add a New Rule

✅ Implement New module

- implements behaviour
- provides business logic for rule

✅ Update the config for the relevant MIX_ENV

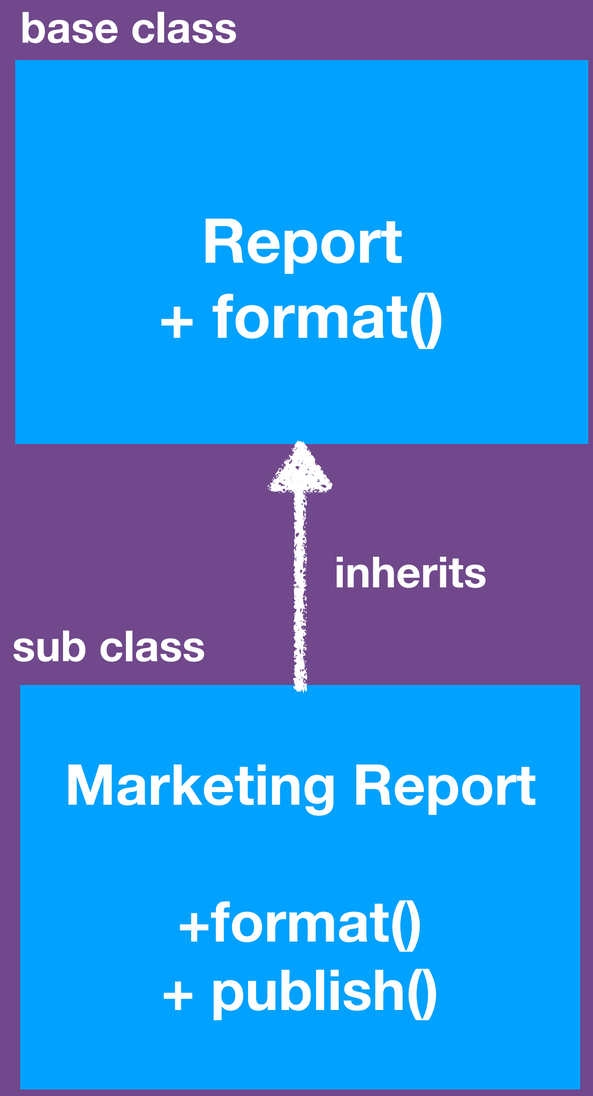
You can add rules without touching any existing code



Liskov Substitution

SOLID

Inheritance



“Liskov's notion of a behavioral subtype defines a notion of substitutability for objects; that is, if *S* is a subtype of *T*, then objects of type *T* in a program may be replaced with objects of type *S* without altering any of the desirable properties of that program.”

– Wikipedia
https://en.wikipedia.org/wiki/Liskov_substitution_principle

OO Substitution

```
MarketingReport marketingReport = new MarketingReport();
marketingReport.format();
```

OO Substitution

Marketing Report

```
MarketingReport marketingReport = new MarketingReport();
marketingReport.format();
```

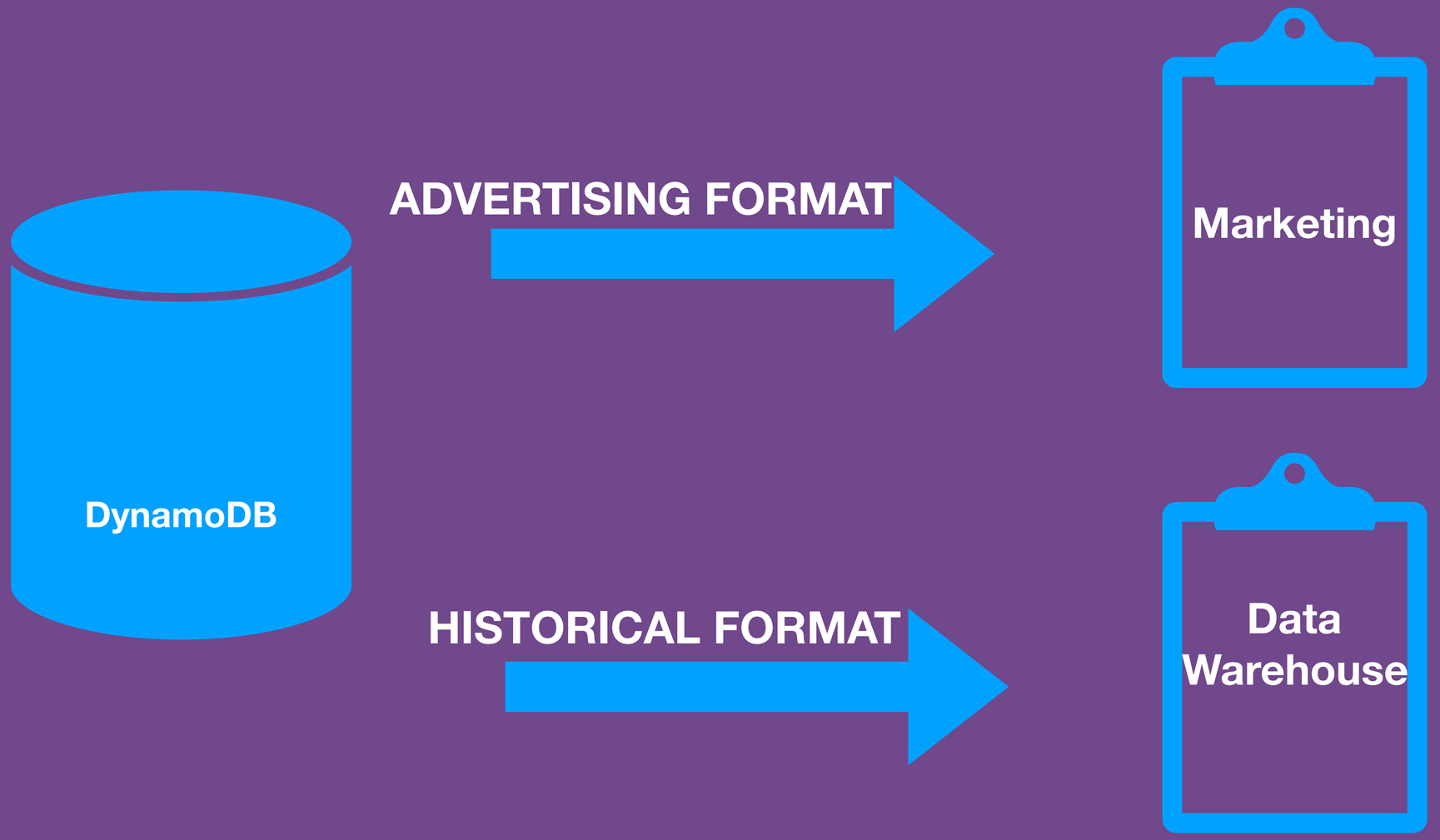
+format()
+ publish()

Report

```
Report marketingReport = new MarketingReport();
marketingReport.format();
```

+ format()

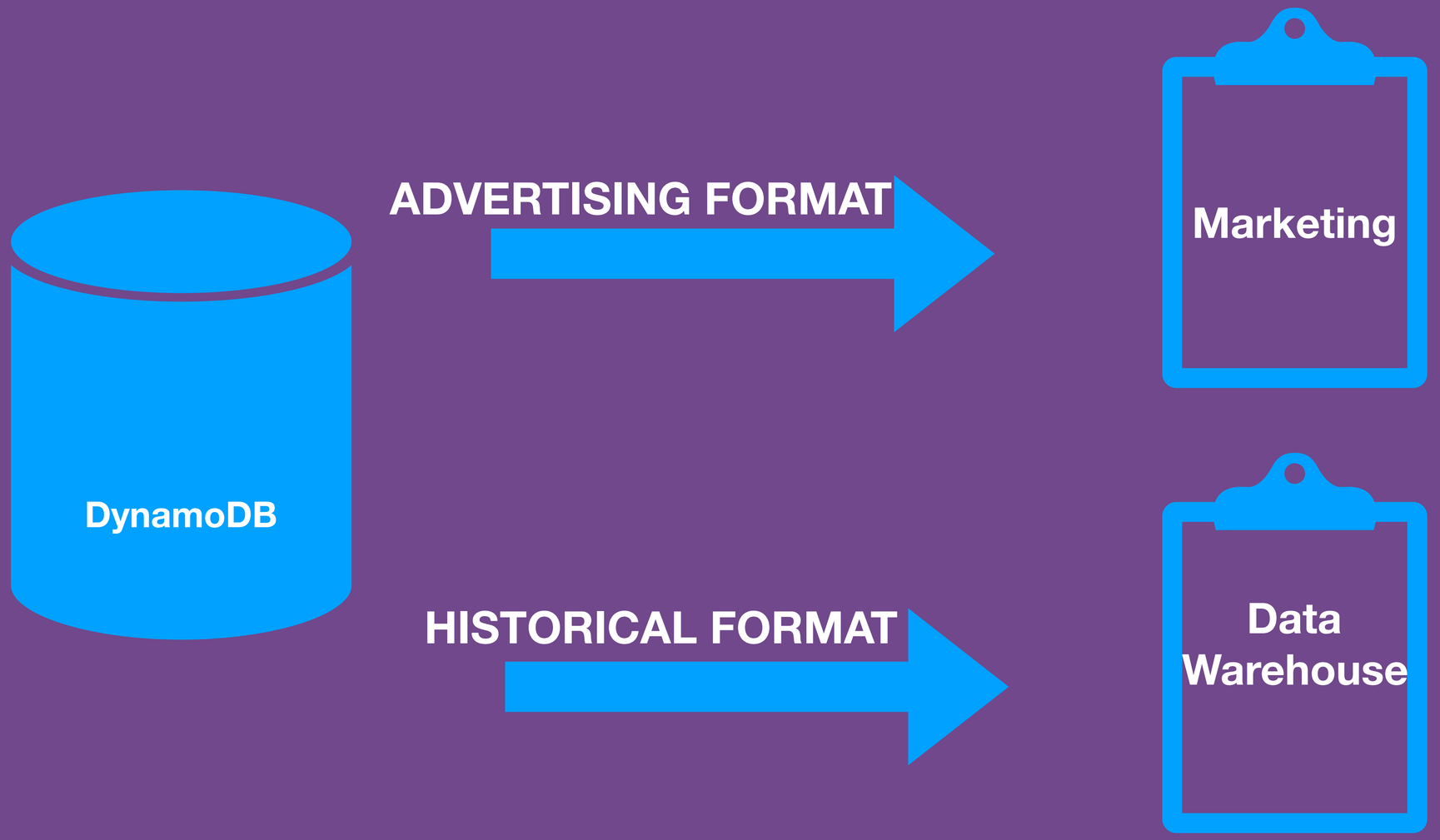
Reporting Capabilities



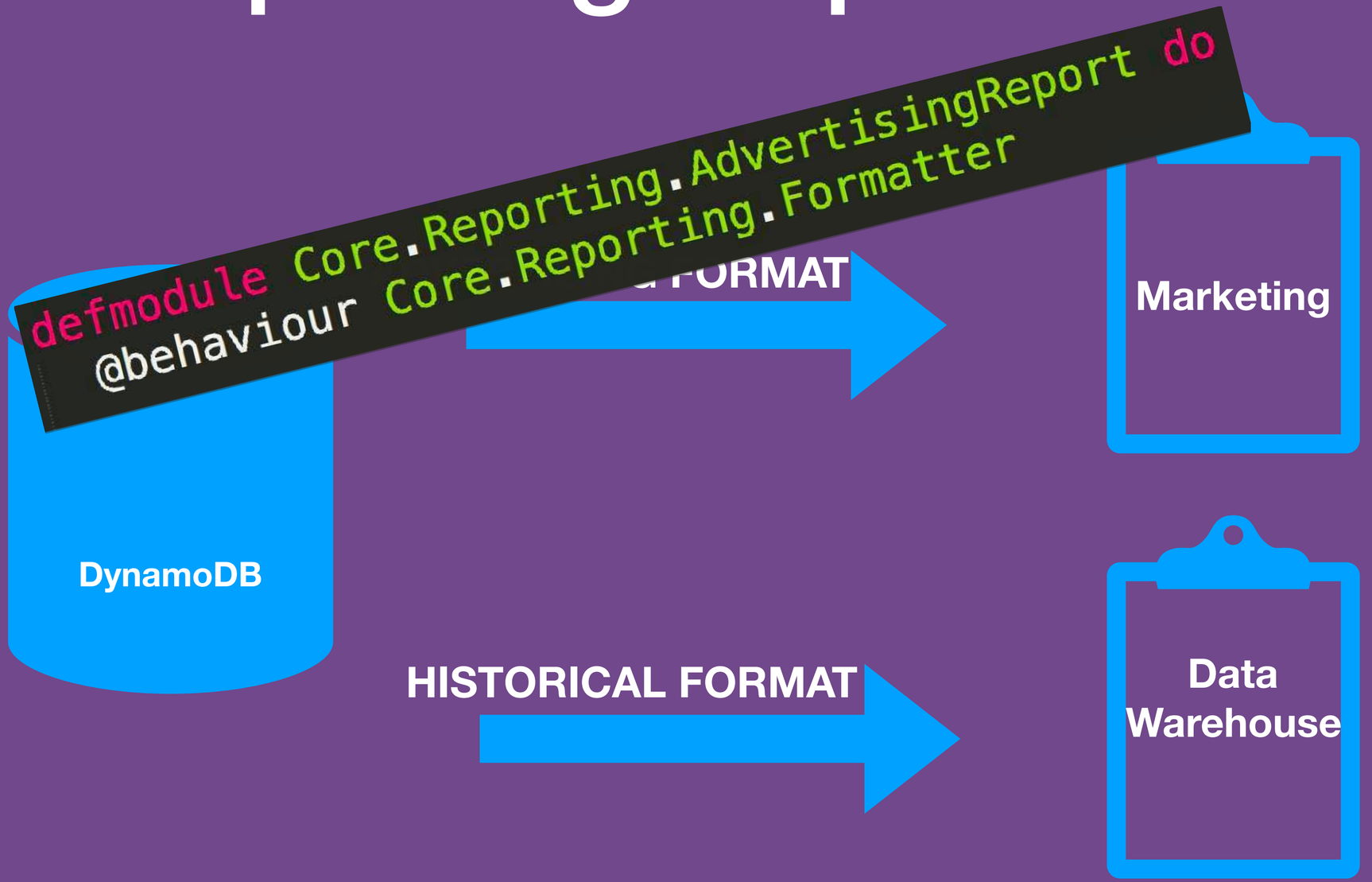
Format Behaviour

```
defmodule Core.Reporting.Formatter do
    @callback format_to_rows(map) ::     {:ok, map}
                                       | {:error, String.t}
end
```

Reporting Capabilities



Reporting Capabilities

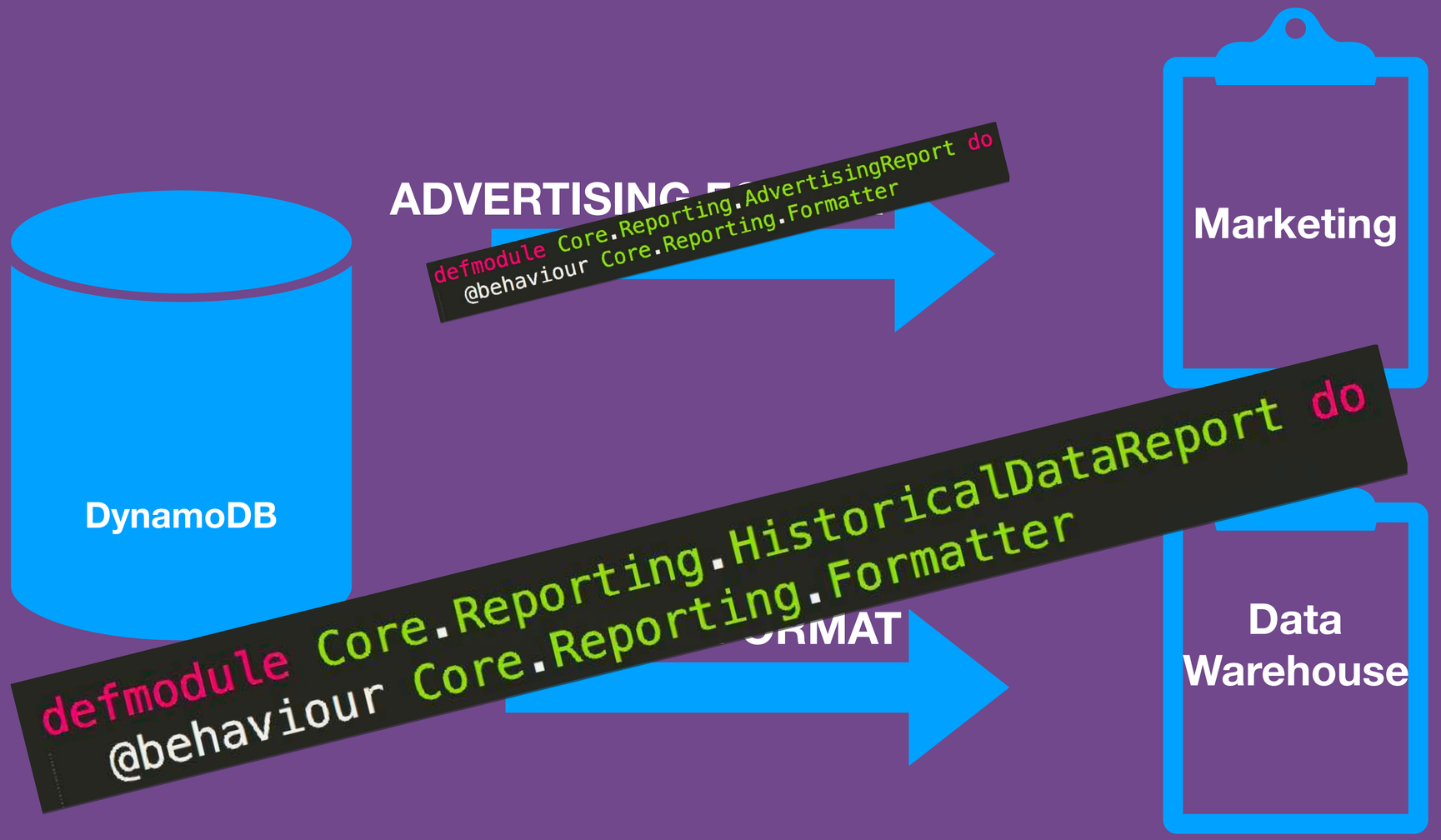


Reports

```elixir
defmodule Core.Reporting.AdvertisingReport do
  @behaviour Core.Reporting.Formatter

  @impl true
  def format_to_rows(data) do
    formatted_data = data
    IO.inspect("format for advertising")
    {:ok, formatted_data}
  end
end
```

Reporting Capabilities



Reporting Capabilities

Generation

```
@spec generate_report(data :: map, formats :: list) :: atom
def generate_report(data, formats) do
```

Generation

```
formatted_reports = Enum.map(formats, fn(format) ->
    format.format_to_rows(data)
end)
```

Liskov Substitution

✅ Add any new module which implements the Behaviour without altering any desirable properties

Legal Report

Legal Report

- Format as per HistoricalDataReport
- Attach a disclaimer
- Attach a header
- Apply colour to headers

Legal Report

```elixir
defmodule Core.Reporting.LegalReportGenerator do

  def generate_report(data, formatter) do
    formatter.format_to_rows(data)
    |> formatter.add_disclaimer
    |> formatter.add_header
    |> formatter.colour
    |> dispatch()
    :ok
  end

  def dispatch(report) do
    IO.inspect("send to external systems")
    :ok
  end
end
```

```elixir
@impl true
def format_to_rows(data) do
```

```
def add_disclaimer(data) do
```

```
def add_header(data) do
```

```
def colour(data) do
```

Generate Legal Report

```
def generate_report(data, formatter) do
 formatter.format_to_rows(data)
  |> formatter.add_disclaimer
  |> formatter.add_header
  |> formatter.colour
  |> dispatch()
 :ok
end
```

Generate Legal Report

```
iex(2)> Core.Reporting.LegalReportGenerator.generate_report(%{"preferences" => "..."},
 Core.Reporting.HistoricalDataReport)
"format for historical data"
"..adding disclaimer.."
"..adding header.."
"..adding colour.."
"send to external systems"
:ok
```

What happens if you generate a Legal Report with another Formatter implementation?

Generate Legal Report

```
def generate_report(data, formatter) do
 formatter.format_to_rows(data)
  |> formatter.add_disclaimer
  |> formatter.add_header
  |> formatter.colour
  |> dispatch()
 :ok
end
```

Generate Legal Report

```
iex(1)> Core.Reporting.LegalReportGenerator.generate_report(%{"preferences" => "..."},
 Core.Reporting.AdvertisingReport)
"format for advertising"
** (UndefinedFunctionError) function Core.Reporting.AdvertisingReport.add_disclaimer/1
 is undefined or private
    (customer_preference_api) Core.Reporting.AdvertisingReport.add_disclaimer({:ok, %{
"preferences" => "..."}})
    (customer_preference_api) lib/core/reporting/legal_report_generator.ex:11: Core.Re
porting.LegalReportGenerator.generate_report/2
iex(1)>
```

Liskov Substitution Violation

According to Liskov Substitution we should be able to plug any implementation in without altering any of the desirable properties

Liskov Substitution

✅ Format as HistoricalDataReport

🤔 Extra Processing to add headers, colours etc

💡 Create a new module that uses the HistoricalDataReport and does a bit more

```
@behaviour Core.Reporting.Formatter
```

```
Core.Reporting.HistoricalDataReport.format_to_rows(data)
```

```
    |> add_disclaimer()
    |> add_header()
    |> colour()
```

```
defmodule Core.Reporting.ReportGenerator do
```

```
defmodule Core.Reporting.AdvertisingReport do
  @behaviour Core.Reporting.Formatter
```

```
defmodule Core.Reporting.HistoricalDataReport do
  @behaviour Core.Reporting.Formatter
```

```
defmodule Core.Reporting.HistoricalDataDecorator do
  @behaviour Core.Reporting.Formatter
```

Liskov Substitution

✅ Wrapping an existing Formatter allowed us to extend it’s functionality

✅ Substitute any implementation safely

Liskov Substitution

Where we have code that expects a generic type (i.e: Behaviour) - Ensure you are only using function calls defined on that Behaviour

But why didn’t we just enhance the original Behaviour with the new functions?

Update Each Implementation

✅ Update HistoricalDataReport

? Update AdvertisingReport

```elixir
defmodule Core.Reporting.AdvertisingReport do
  @behaviour Core.Reporting.Formatter

  @impl true
  def format_to_rows(data) do
    formatted_data = data
    IO.inspect("format for advertising")
    #... some data clensing and reformatting logic
    {:ok, formatted_data}
  end

  def add_disclaimer(data) do
    IO.puts("nothing to do..")
    data
  end

  def add_header(data) do
    IO.puts("nothing to do..")
    data
  end

  def colour(data) do
    IO.puts("nothing to do..")
    data
  end
end
```

 Bloated the Behaviour to satisfy a new requirement


Interface Segregation
SOLID
@gemcfadyen

"The **interface-segregation principle (ISP)** states that no client should be forced to depend on methods it does not use.[1] ISP splits interfaces that are very large into smaller and more specific ones so that clients will only have to know about the methods that are of interest to them."

– Wikipedia
https://en.wikipedia.org/wiki/Interface_segregation_principle

Interface Segregation Signs

- Look for Behaviours which have methods that don’t apply to all implementors
- Look for Behaviours that have many function definitions

Interface Segregation Signs

```elixir
defmodule Core.Reporting.Formatter do
  @callback format_to_rows([map]) ::  {:ok, [map]}
                                    | {:error, String.t}

  @callback add_disclaimer([map]) :: [map]

  @callback add_header([map]) :: [map]

  @callback colour([map]) :: [map]
end
```

Split Behaviour

```elixir
defmodule Core.Reporting.Formatter do
  @callback format_to_rows([map]) ::  {:ok, [map]}
                                    | {:error, String.t}
end
```

Split Behaviour

```elixir
defmodule Core.Reporting.Formatter do
  @callback format_to_rows([map]) ::  {:ok, [map]}
                                    | {:error, String.t}
end
```

```elixir
defmodule Reporting.Presentation do
  @callback add_disclaimer([map]) :: [map]

  @callback add_header([map]) :: [map]

  @callback colour([map]) :: [map]
end
```

```
@behaviour Core.Reporting.Formatter
@behaviour Core.Reporting.Presentation
```

```
 def generate_report(data, formats) do
 formatted_reports = Enum.map(formats, fn(format) ->
  format.format_to_rows(data)
 end)
 dispatch(formatted_reports)
end
```

```
def present_report(formatted_data, presenters) do
  formatted_reports = Enum.map(presenters,
   fn(presenter) ->
     presenter.add_disclaimer()
     |> presenter.add_header()
     |> presenter.colour()
     |> dispatch
   end)
end
```

Split Processing

```
1  defmodule Core.Reporting.ReportingClient do
2    def entry_point(data) do
3     formatted_data = generate_report(data,
4                                      [Core.Reporting.TrendsReport,
5                                       Core.Reporting.AdvertisingReport])
6                    |> dispatch
7
8     present_report(data,
9                    [Core.Reporting.HistoricalDataReport])
10    |> dispatch
11   end
12 end
```

✅ Force implementations only on those modules that actually need them

Dependency Inversion

SOLID

@gemcfadyen

The dependency inversion principle refers to a specific form of decoupling software modules.

High-level modules should not depend on low-level modules. Both should depend on abstractions.

Abstractions should not depend on details. Details should depend on abstractions.

– Wikipedia
https://en.wikipedia.org/wiki/Dependency_inversion_principle

GET /customers/{id}

```
ExAws.Dynamo.get_item("customer-preferences",
                      %{id: customer_id})
|> ExAws.request
```

Fetch Customer Preferences

```elixir
ExAws.Dynamo.get_item("customer-preferences",
                      %{id: customer_id})
|> ExAws.request
```

```elixir
ExAws.Dynamo.get_item("customer-preferences",
                      %{id: customer_id})
|> ExAws.request
```

FAKE DATABASE

MIX_ENV = test
- fake-database

MIX_ENV = dev
- local-database

MIX_ENV = prod
- aws-database

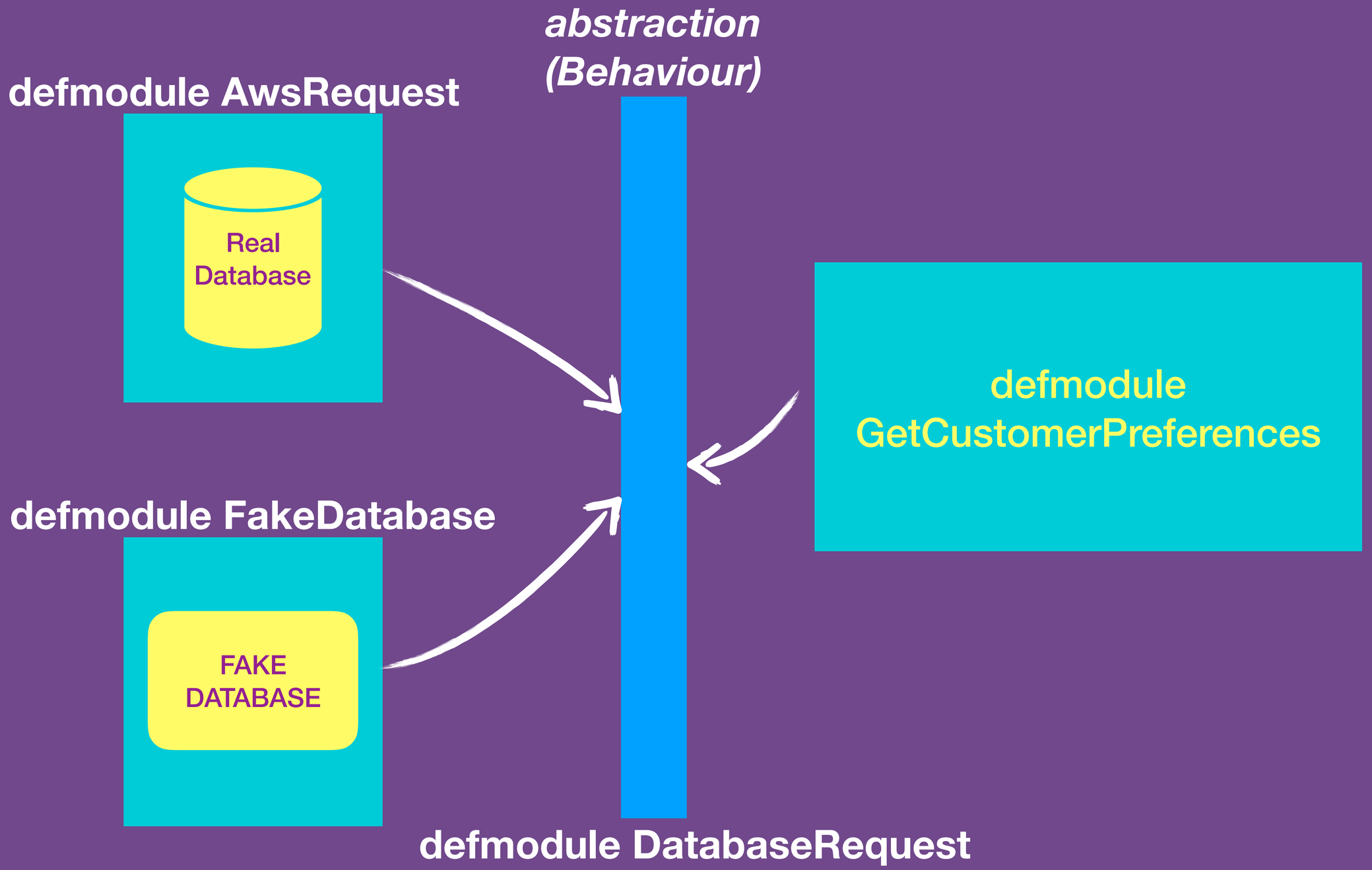
Isolation

```
def for(customer_id) do
 ExAws.Dynamo.get_item("customer-preferences",
                       %{id: customer_id})
 |> ExAws.request
 |> decode
end
```

```
|> ExAws.request
```

Isolation

Use Config to Substitute Test Doubles


abstraction
(Behaviour)
defmodule AwsRequest
Real
Database
defmodule
GetCustomerPreferences
defmodule FakeDatabase
FAKE
DATABASE
defmodule DatabaseRequest

Create Abstraction

```elixir
defmodule Database.Request do
  @callback request(ExAws.Operation.t, Keyword.t) ::  {:ok, term}
                                                    | {:error, term}

  @callback request(ExAws.Operation.t) ::   {:ok, term}
                                          | {:error, term}
end
```

Wrap the real database functionality

```elixir
defmodule Database.AwsRequest do
  @behaviour Database.Request

  @impl true
  def request(op, config_overrides \\ []) do
    ExAws.request(op, config_overrides)
  end
end
```

```elixir
defmodule Database.AwsRequest do
  @behaviour Database.Request

  @impl true
  def request(op, config_overrides \\ []) do
    ExAws.request(op, config_overrides)
  end
end
```

Provide a fake database request module

```elixir
defmodule Database.FakeDatabase do
  @behaviour Database.Request

  @impl true
  def request(op, config_overrides \\ [])

  @impl true
  def request(%{http_method: :post}, _) do
   {:ok, canned_result()}
  end

  def canned_result do
   IO.puts("Saving data...")

   %{"Item" =>
    %{"id" => %{"S" => "uuid-1"},
     "preferences" => %{"M" =>
      %{"accommodation" => %{"M" =>
       %{"apartment" => %{"M" =>
        %{"bedrooms" => %{"N" => "2"},
          "catering" => %{"S" => "self_catering"},
          "parking" => %{"S" => "secure"}}
          }}}}}}}
  end
end
```

MIX_ENV=test fake_database.ex

MIX_ENV=prod aws_request.ex

customer_preference_api/config/test.exs

```
config :customer_preference_api,
  aws_request: Database.FakeDatabase
```

customer_preference_api/config/prod.exs

```
config :customer_preference_api,
  aws_request: Database.AwsRequest
```

Lookup implementation

```
database = Application.get_env(:customer_preference_api,
                               :aws_request)
```

Configure different Scenarios

- Testing scenarios where nothing is found
- Testing scenarios where multiple entries found….

Configure Fake via Config

✅ Isolates ‘fake’ functionality to specific MIX_ENV

❌ Messy to handle different canned results for different scenarios

❌ Difficult to know what ‘fake’ logic is being executed for what test

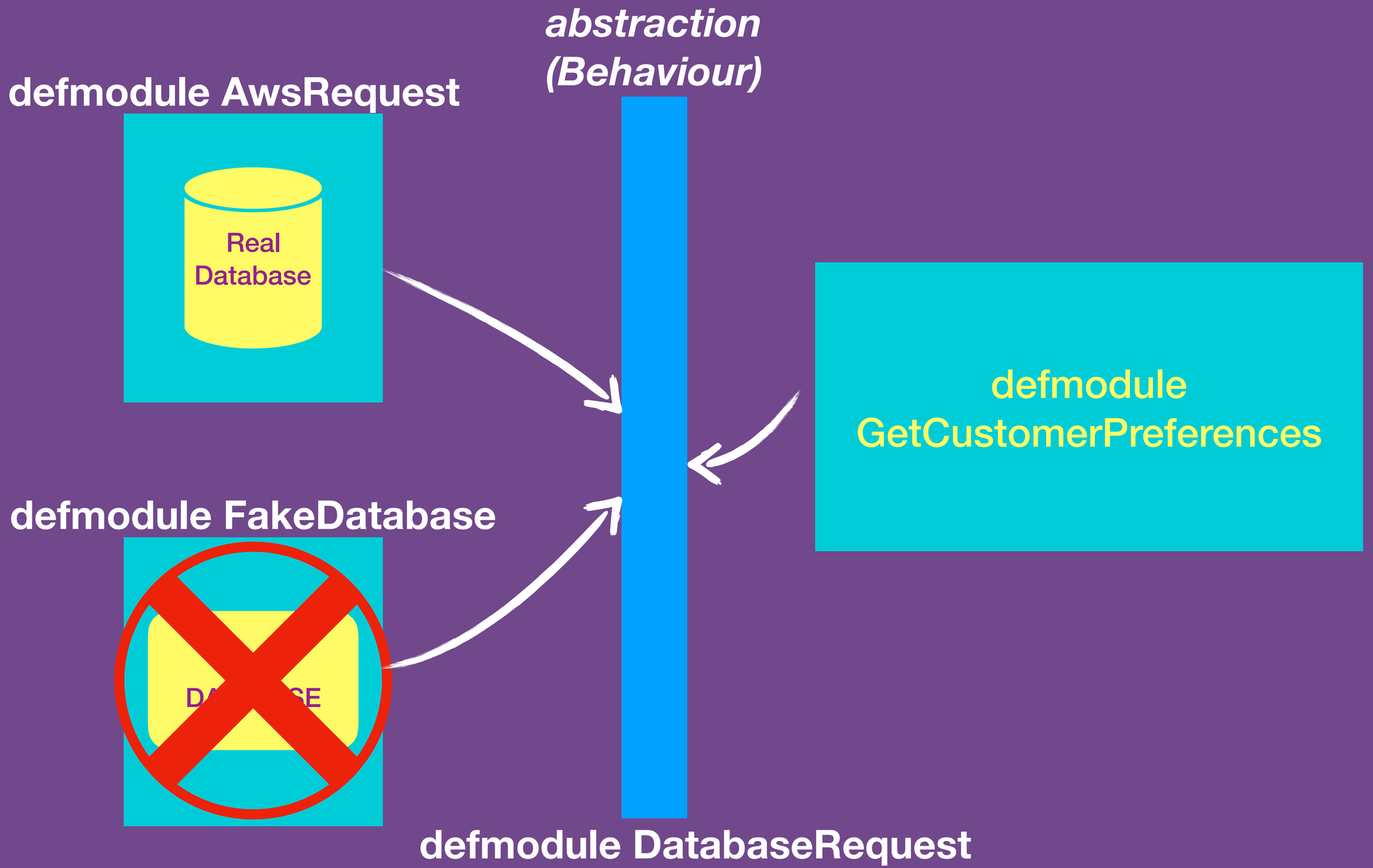
❌ Fake implementations can diverge from the real implementations

Mocking Library

https://github.com/plataformatec/mox

Add dependency

```elixir
defp deps do
  [
    ...
    {:mox, "~> 0.3.1"},
    ...
  ]
end
```


abstraction
(Behaviour)
defmodule AwsRequest
Real
Database
defmodule
GetCustomerPreferences
defmodule FakeDatabase
defmodule DatabaseRequest

Wrap ‘real’ implementation

```elixir
defmodule Database.AwsRequest do
  @behaviour Database.Request

  @impl true
  def request(op, config_overrides \\ []) do
    ExAws.request(op, config_overrides)
  end
end
```

Configure Implementations - prod config

```
1  config :customer_preference_api,
2   aws_request: Database.AwsRequest
```

Configure Implementations - test config

```
1  config :customer_preference_api,
2   aws_request: Database.MockRequest,
```

Test Expectations

```
1 ExUnit.start()
2
3 Mox.defmock(Database.MockRequest, for: Database.Request)
```

Configure Implementations

```
1  defmodule Core.Preferences.GetCustomerPreferences do
2   @aws_request Application.get_env(:customer_preference_api,
3                                    :aws_request)
4
5   def for(customer_id) do
6    ExAws.Dynamo.get_item("customer-preferences",
7                          %{id: customer_id})
8    |> @aws_request.request
9    |> decode
10  end
11  ...
12 end
```

Test Expectations

```
import Mox
@mock_request Database.MockRequest
```

Test Expectations

```
expect(@mock_request,
       :request,
       fn _ -> {:ok, canned_result()} end)
```

Define expectation

Test Expectations

```
  1) test fetches preferences from database (Core.Preferences.GetCustomerPrefere
ncesTest)
     test/core/preferences/get_customer_preferences_test.exs:7
     ** (Mox.UnexpectedCallError) no expectation defined for Database.MockReques
t.request/1 in process #PID<0.591.0>
     code: {:ok, result} = Core.Preferences.GetCustomerPreferences.for("1")
     stacktrace:
       (mox) lib/mox.ex:438: Mox.__dispatch__/4
       (customer_preference_api) lib/core/preferences/get_customer_preferences.e
x:6: Core.Preferences.GetCustomerPreferences.for/1
       test/core/preferences/get_customer_preferences_test.exs:9: (test)
```

Mocking Libraries

✅ Easy to setup different scenarios

✅ Fake behaviour lives with the test

✅ Don’t have to maintain any ‘fake’ implementations

❌ Mix spits out some warnings about the mock modules not physically existing

Dependency Inversion

SOLID

Depend on abstractions

Provide to the application, a specific implementation of a dependency to use depending on the circumstance


Can
you
write
SOLID
Elixir?
@gemcfadyen


Can you write SOLID Elixir?
yes!
@gemcfadyen

Does it make for idiomatic Elixir?

Do we need a set of functional design principles?

SOLID ✅

Functions

SOLID ✅

Higher Order Functions

SOLID

Pure Functions & Referential Transparency

Pure Functions

Deterministic with no side effects

input -> output

```
def function(input) do
  output
end
```

Pure Functions

```
def add(number, number) do
  number + number
end
```

Pure Functions

```
def lucky_number() do
  add(2, 2)
  |> add(2)
  |> add(1)
end
```

Referential Transparency

SOLID ✅

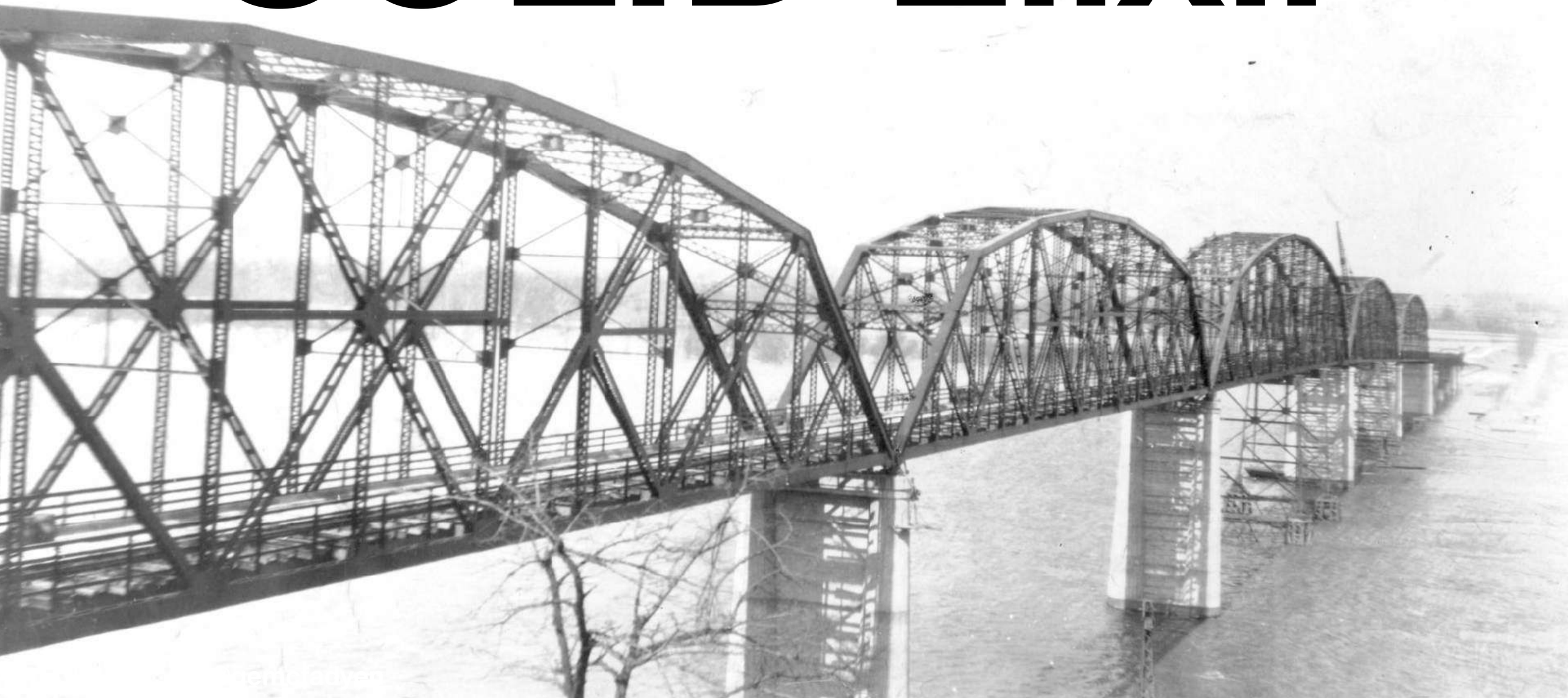
Behaviour

SOLID ✅

Abstractions

Immutability
Recursion
Pattern Matching
Currying
...

SOLID Elixir

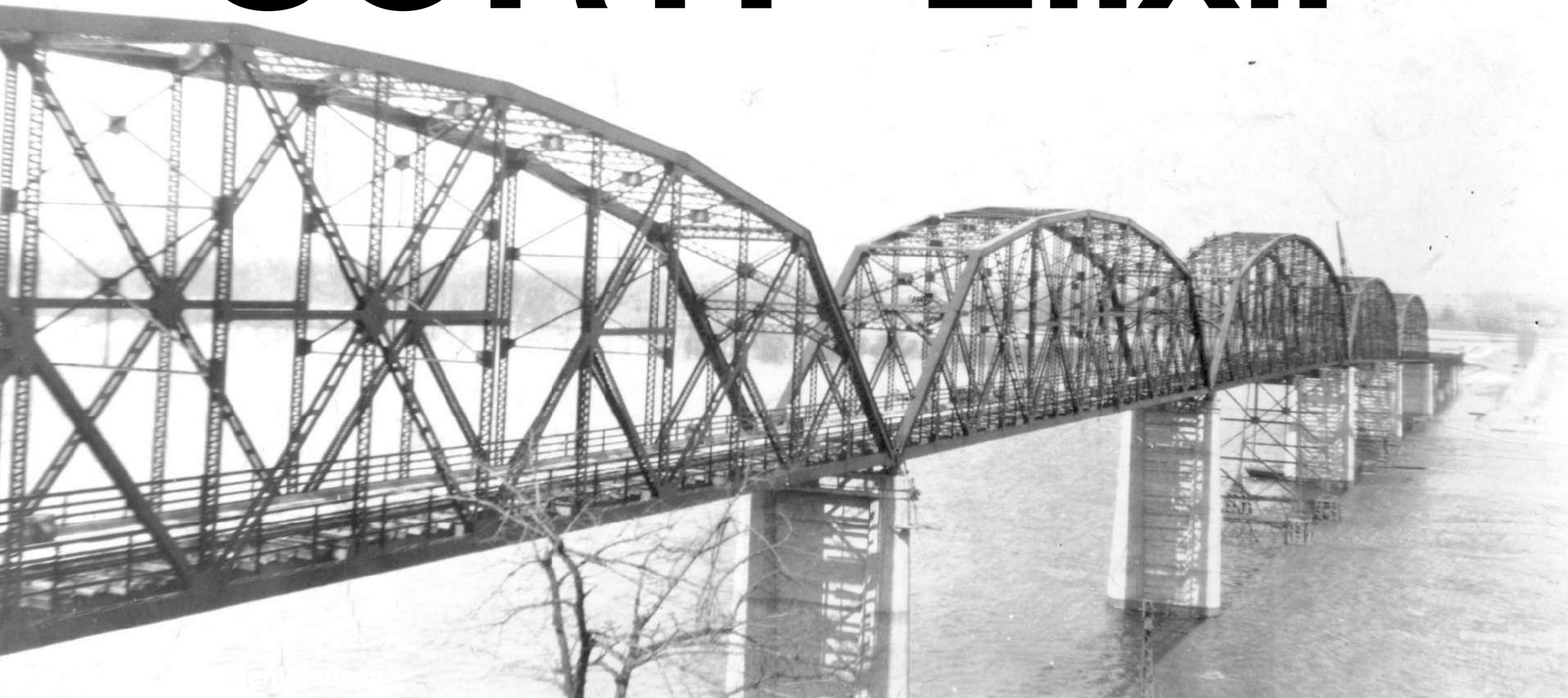


S- - - - Elixir

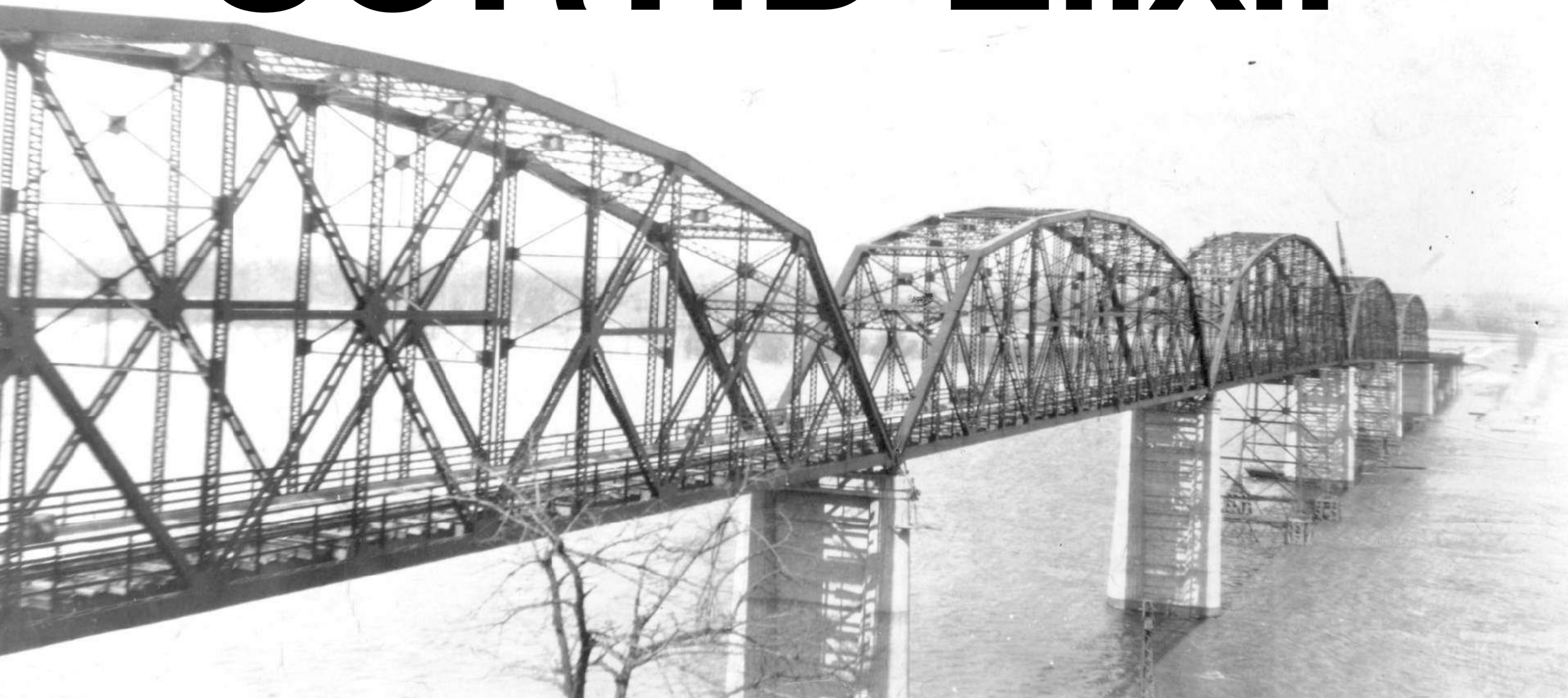
SO- - - Elixir

SORT- - Elixir

SORTI- Elixir



SORTID Elixir



Thank-you!